



Operations Research

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Exercise Sheet 3

Demo Exercises

Exercise D3.1. Revised Simplex

Solve the following LP using the revised simplex algorithm. Start with the feasible basis (x_1, x_6) .

$$\begin{aligned} \text{Max} \quad & 12x_1 + 20x_2 + 16x_3 + 12x_4 \\ \text{s.t.} \quad & 4x_1 + 8x_2 + 2x_3 \leq 400 \\ & 2x_1 + 5x_2 + 10x_3 + 4x_4 \leq 400 \\ & x \geq 0 \end{aligned}$$

Exercise D3.2. Waldgeist Distillery

The Waldgeist distillery, led by Steffen, produces two different spirits from raspberries, blueberries and strawberries: the Himmelrot liqueur and the Nachtblau fruit brandy. The available capacities, the required quantity of raspberries and blueberries in *kg* as well as the unit price and costs per bottle of each variety can be seen in the following table:

Resources	Himmelrot	Nachtblau	Capacity
Raspberry	2	1	20
Blueberry	2	3	18
Strawberry	1	1/2	8
Unit price	15	20	
Unit cost	10	16	

Georg, an employee at Waldgeist, has already modeled the above production problem as a linear program (P) . By using the Simplex algorithm he found the following optimal table. Unfortunately, Georg was too conscientious in the quality control of Nachtblau and spilled some of the good wine on his invoice, which made some entries unreadable.

$$\begin{aligned} (P) \quad \text{Max} \quad & 5x_1 + 4x_2 \\ \text{s.t.} \quad & 2x_1 + x_2 \leq 20 \\ & 2x_1 + 3x_2 \leq 18 \\ & x_1 + 1/2x_2 \leq 8 \\ & x \in \mathbb{R}_{\geq 0}^2 \end{aligned}$$

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
Z			0	0.75	3.5	
			1	0	-2	
x_2			0	0.5	-1	
x_1			0	-0.25	1.5	7.5

- a) Explain the meaning of the objective function, the constraints and the decision variables of (P)
- b) Complete the final table above for Georg.
- c) What is the maximum amount that Steffen should pay for $1kg$ of additional raspberries? Justify your answer.
- d) Calculate the limits for the selling price of Nachtblau within which the optimal basis does not change. Explain in detail how the optimal production schedule and resource stocks change, if the selling price violates these limits.
- e) The supply of strawberries and blueberries varies greatly.
 - (i) Within what limits can the blueberry capacity move so that both products can continue to be produced?
 - (ii) Within what limits can the strawberry capacity move so that both products can continue to be produced?
 - (iii) Determine the respective shadow prices and reduced costs if the strawberry inventory violates these limits.
- f) Steffen has the opportunity to exchange strawberries and blueberries for each other at the market. For q kilograms of one type of berry, he receives q kilograms of the other type of berry. Calculate the best split of strawberries and blueberries while maintaining the current base of the optimal table.
- g) A new product *Fruchttraum* contains $4kg$ raspberries, $2kg$ blueberries and $3kg$ strawberries. Furthermore, *Fruchttraum* generates costs of 13 per bottle.
 - (i) What is the minimum bottle price of *Fruchttraum* that must be in order for it to be included in the optimal production program?
 - (ii) What happens if *Fruchttraum* needs $6kg$ of strawberries per bottle?
- h) Steffen points out to Georg that only whole (and full!) bottles of Nachtblau and Himmelrot can be produced and therefore the solution calculated by Georg cannot be implemented.
 - (i) How does this change the problem (P) ?
 - (ii) What does the solution calculated by Georg tell you about the achievable profit of the adapted problem?
- i) *Self-study*: Solve the above subtasks using Gurobi.